

Disproof of Conjecture 00000007674

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The definition in the problem is

$$\kappa = \limsup_{n \rightarrow \infty} \frac{1}{n^2} \log d_n,$$

where d_n is a pole order. Whenever this displayed real-valued definition is meaningful, the relevant pole orders are positive integers. Hence $d_n \geq 1$, so $\log d_n \geq 0$ and every term in the limsup is nonnegative. Therefore

$$\kappa \geq 0.$$

Now choose the algebraic number $q = 1/2$, which satisfies the stated hypothesis $0 < |q| < 1$. The conjecture asserts

$$\kappa = \log |q| = \log(1/2) < 0,$$

contradicting $\kappa \geq 0$. Thus the universal claim is false. Notice also that if a “pole order” $d_n = 0$ were permitted, the real logarithm in the definition would be undefined; that convention cannot rescue the asserted real equality.

The Lean development works in an arbitrary linearly ordered codomain with a strictly increasing logarithm satisfying $\log(1) = 0$. It kernel-checks $1/2 < 1$, derives $\log(1/2) < 0$ from strict monotonicity, and proves that no nonnegative value can equal this negative logarithm. Thus it formalizes the entire order-theoretic core without adding analytic axioms.